MGTE 31423
Advanced Optimization Methods in Management Science

3. Game Theory – Part 1

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❖ Objectives

- Be able to structure a game in normal form

- Be able to identify a Nash equilibrium

- Be able to identify the best strategy for the player and for the opponent

Outline of the session

The strategies the player and opponent will be using is called either PURE strategy or MIXED strategies

- Pure Strategy games
  - Minimax criteria
  - Maximin Criteria
  - Value of the game
  - Dominant strategies
  - Equilibrium
- Mixed Strategy games
  - No equilibrium
  - Mixed optimal solution
  - LP formulation

- **pure strategy: only one particular strategy is played**
- **mixed strategy: more than one strategy is used**

Payoff table

Example 1:

	S_1	S_2	
d_1	\$200,000	\$-20,000	
d_2	\$150,000	\$ 20,000	
d_3	\$100,000	\$ 60,000	

d_1- lease the large computer system
d_2- lease the medium size computer system
d_3- lease the small computer system

s_1- high customer acceptances of the company service
s_2- low customer acceptances of the company service

Elements of Game Theory

- Competition between multiple players
  (only focus on 2 players)
- Explicit objective function
- Determine strategy to optimize objective
- Types of games
  - 2*2 strategies, 2*n strategies, m*2 strategies and m*n strategies

- Two-person, **zero-sum** games most common

- Decision theory helps us in identifying the best decision alternative to a given problem. The criterion used in selecting the best decision is the pay- off.

- Game theory is an extension to decision theory where the states of nature includes decision taken by parties interested in our decision as well as parties not interested in our discussion.

- The origin of this theory relates to games with economic gains and that is why it is called game theory. However, it has proved its value in business decision making.

- Another feature in game theory is that it deals with a sequence of repeated decisions rather than a single decision that is taken.
- Solution to game theory problems **are based on pessimistic** approach (player→MAX – MIN, OR opponent→MIN – MAX criteria) of decision theory.

Terminology in Game Theory

- **PLAYER** :- Each interested party in the game is a player

  Considered 2 player game

- **STRATERGY** :- The decision rule by which a player determines his course of action is called strategy

  a) pure strategy: only one particular strategy is played

  b) mixed strategy : more than one strategy is used

- **Value of Game** :- This is the average amount that one will win in the long run if all participants use their best strategies. If the game has a saddle point then the saddle point is the value of the game

- **A "saddle-point" in a two-person constant-sum game is the outcome that rational players would choose. Saddle Point: Maximin value =minimax value**

Terminology in Game Theory

- **Payoff table** : In game theory, a payoff matrix is a table in which strategies of one player are listed in rows and those of the other player in columns and the cells show payoffs to each player such that the payoff of the row player is listed first.

		Opponent	
		B1	B2
Player	A1	7	8
	A2	4	-5
	A3	3	1

If A3 and B1 is selected player A get 3 and player B(opponent) loose 3s

If A2 and B2 is selected player A looses 5 and player B get 5

Terminology in Game Theory

- **Principal criterion:**

  a) Maximin - maximise the minimum guaranteed gain of player A

  b) Minimax – minimise the maximum lose of player B

Saddle Point: Maximin value =minimax value

2 person Zero sum game: In a game with 2 players, if the gain of one player is equal to the loss of another player (opponent), then that game is called 2 person zero sum game

		Opponent	
		B1	B2
Player	A1	7	8
	A2	4	-5
	A3	3	1

		Opponent	
		B1	B2
Player	A1	7,-7	8,-8
	A2	4,-4	-5,5
	A3	3,-3	1,-1

2 person zero sum game

- We shall be dealing with the simplest form of Game Theory, namely, Two person Zero sum game.
- In general, a game may include many persons or parties. However, what we hope to study will **have only two persons**. One is called the **Player,** and the other is called the **Opponent**.
- Total gain of both the player and the opponent is zero. That is what the player gains and the opponent loses. This is why it is called a two person zero sum game.
- The decision taken by either the player or the opponent is not a single decision. Instead, it is a series of decisions taken in rotation. The player or the opponent may decide to make decisions in a given ratio.
- The strategies the player and opponent will be using are called either PURE strategy or MIXED strategies

There are six properties necessary in order that game theory may be applied.

They are

- The number of participants should be finite (usually 2)
- Each participant should have a finite number of courses of action or decision
- The participants that wish to apply game theory must know all the courses of action available to other parties
- The gains of the participants should depend on the actions of others as well as himself.
- These gains of the participants should be finite.
- It should be possible to evaluate the gains of the participants

Situations that satisfy the above condition is called a **GAME**.

There are two types of strategies: Namely

t of

Pure Strategy

Pure strategy is when the player or opponent decides on the same decision at every round.

Mixed Strategy

When the player or opponent mixes the decisions in a given ratio, we call it a mix strategy. For example, the player may have three decision alternatives and he may want to take the series of decisions in the ratio 1:3:2.

That is when he decides on decision 1 once he would decide on decision 2 thrice and decision 3 twice.

This ratio 1:3:2 is called his strategy. Or [1/6, 3/6, 2/6] probability

The Gain Matrix:

The gain matrix is a table that tells us the value received by the player when the player and the opponent decide on a set of decisions.
The following table is an example of a gain matrix.

Player (offensive player)	Opponent (defensive player)	
	Decision 1	Decision 2
Decision 1	7	8
Decision 2	4	5
Decision 3	3	1

Value that opponent should give to the player

For example, when player decides on Decision (2) and the opponent decides on Decision (2) the value received by player is 5.

Value of Game:

The value received by the player in the long run is called the value of the game.

SADDLE POINTS

- This occurs when the gain of the best decision for the player is equal to the gain of the best decision of the opponent.

- The best decision is based on the **Pessimistic Approach**. That is selecting the best among the worst that could happen from each decision.

- For the player it is the MAX-MIN criteria and
- For opponent it is the MIN-MAX criteria in general.

Maximin - maximise the minimum guaranteed gain of player A

Minimax – minimise the maximum loss of player B

Saddle point

For the player it is the MAX-MIN criteria and
for opponent it is the MIN-MAX criteria in general.

Saddle point

		OPPONENT			Worst that could happen to player
		B1	B2	B3	
PLAYER	A1	3	-1	2	-1
	A2	-4	-1	13	-4
	A3	2	-2	-1	-2
Worst for Opponent		3	-1	13	

- When, as in the above example the best value for player and opponent are equal, we say that **saddle point** exists.

  A "saddle-point" in a two-person constant-sum game is the outcome that rational players would choose. Saddle Point: Maximin value =minimax value

- When saddle point exists, the best strategy is a pure strategy.
- The pure strategy of player is to keep on playing A2. We can denote this as 0:1:0
- The value of the game is equal to the gain of the best decision of player or opponent which is -1.
- (The value of the game is also the value at the intersection of row A1 and column B2 namely -1)
- When saddle points exits the best for both the player as well as the opponent is to play their best decisions. It is assumed that the opponent will be playing his best decision and consequently best for the player is to play his best decision which results in a pure strategy.

Prepare the pay off table of opponent

		OPPONENT			Worst that could happen to player
		B1	B2	B3	
PLAYER	A1	3	-1	2	-1
	A2	-4	-1	13	-4
	A3	2	-2	-1	-2
Worst for Opponent		3	-1	13	

	B1	B2	B3
A1	3,-3	-1, 1	2,-2
A2	-4,4	-1, 1	13, -13
A3	2, -2	-2,2	-1, 1

	A1	A2	A3
B1	-3	4	-2
B2	1	1	2
B3	-2	-13	1

	B1	B2	B3
A1	3,-3	-1, 1	2,-2
A2	-4,4	-1, 1	13, -13
A3	2, -2	-2,2	-1, 1

	B1	B2	B3	
A1	3	-1	2	-1
A2	-4	-1	13	-4
A3	2	-2	-1	-2
	3	-1	13	

	A1	A2	A3	
B1	-3	4	-2	-3
B2	1	1	2	1
B3	-2	-13	1	-13
	1	4	2	

An example of a pure strategy game

		opponent			
		B1	B2	row min	
player	A1	7	8	7	maximin value
	A2	4	-5	-5	
	A3	3	1	1	
	column max	7	8		
		Minimax value			

Saddle point: when Maximin value = Minimax value

7 = 7

Value of the game is **7** and it is the value of the saddle point

A[1,0,0] **B[1,0]**

p q r

$P(A_1)$ $P(A_2)$ $P(A_2)$

When we do not have a saddle point in our game
Then we use mixed strategies

An example of a mixed strategy game

	Opponent				
	B1	B2	B3	B4	Min gain
A1	5	4	3	7	3
A2	3	2	1	4	1
A3	9	1	2	8	1
A4	7	2	10	3	2
Max Loss	9	4	10	8	

Maximize the minimum gain: MaxMin = 3

Minimize the maximum loss: MinMax = 4

Since Maximin value ≠ Minimax value, a saddle point does not exist

$3 \neq 4$

$3 \leq$ Value of the game ≤ 4

Since we do not have a saddle point in our game, we use mixed strategies.

Eliminating redundancies in the payoff table

	Opponent			
	B1	B2	B3	B4
A1	5	4	3	7
A2	3	2	1	4
A3	9	1	2	8
A4	7	2	10	3

Eliminating Redundancies in the payoff table using the Rule for Dominance (Iterated elimination of strictly dominated strategies: IESDS)

a. If **all the elements in a row are greater than or equal** to the corresponding elements in a second row, then the second row is dominated and is deleted from the matrix.

b. If **all the elements in a column are less than or equal** to the corresponding elements in a second column, then the second column is dominated and is deleted from the matrix.

Procedure in solving game theory problems

1: Check for dominance and reduce matrixes
2: Check for saddle points.
3: If player has only two decisions, then use oddment or graphical method to solve.
4: If player has more than two decisions then use simplex method to solve.

	Opponent				
	B1	B2	B3	B4	Total gain
A1	5	4	3	7	19
A2	3	2	1	4	10 Min
A3	9	1	2	8	20
A4	7	2	10	3	22 Max
Total Loss	24	9	16	22	
	Max		Min		

A4 – A2
A4 – A1
A4 – A3

A3 – A1
A3 – A2
A1 – A2

Dominance:

		Opponent			
		B1	B2	B3	B4
PLAYER	A1	5	4	3	7
	A2	3	2	1	4
	A3	9	1	2	8
	A4	7	2	10	3

A2 is redundant

		opponent	
		B2	B3
PLAYER	A1	4	3
	A3	1	2
	A4	2	10

- Ex (II): Check for dominance and saddle points in respect of the following gain matrix.

Ex 2.1		Opponent			
		B1	B2	B3	B4
PLAYER	A1	7	2	8	5
	A2	9	5	3	1
	A3	8	3	6	10
	A4	3	1	4	3

Exercise 2.2		Opponent				
		B1	B2	B3	B4	B5
PLAYER	A1	4	6	5	10	6
	A2	7	8	5	9	10
	A3	8	9	11	10	9
	A4	6	4	10	6	4

		Opponent					
		B1	B2	B3	B4	row min	tot
PLAYER	A1	7	2	8	5	2	22
	A2	9	5	3	1	1	18
	A3	8	3	6	10	3	27
	A4	3	1	4	3	1	11
	col max	9	5	8	10		
	tot	27	11	21	19		

		Opponent					
		B1	B2	B3	B4	row min	tot
PLAYER	A1	7	2	8	5	2	22
	A2	9	5	3	1	1	18
	A3	8	3	6	10	3	27
	A4	3	1	4	3	1	11
	col max	9	5	8	10		
	tot	27	11	21	19		

		Opponent		
		B2	B3	B4
PLAYER	A1	2	8	5
	A2	5	3	1
	A3	3	6	10

<u>Player dominance</u>: - Decision A4 is dominated by A3 and decision A4 is deleted

<u>Opponents' dominance</u>: - Opponents' decision B1 is dominated by B2 and decision B1 is deleted.

The reduced gain matrix is as follows

For example, redundant constraints in linear programming problems

$$\text{Min } 4X + 5Y$$

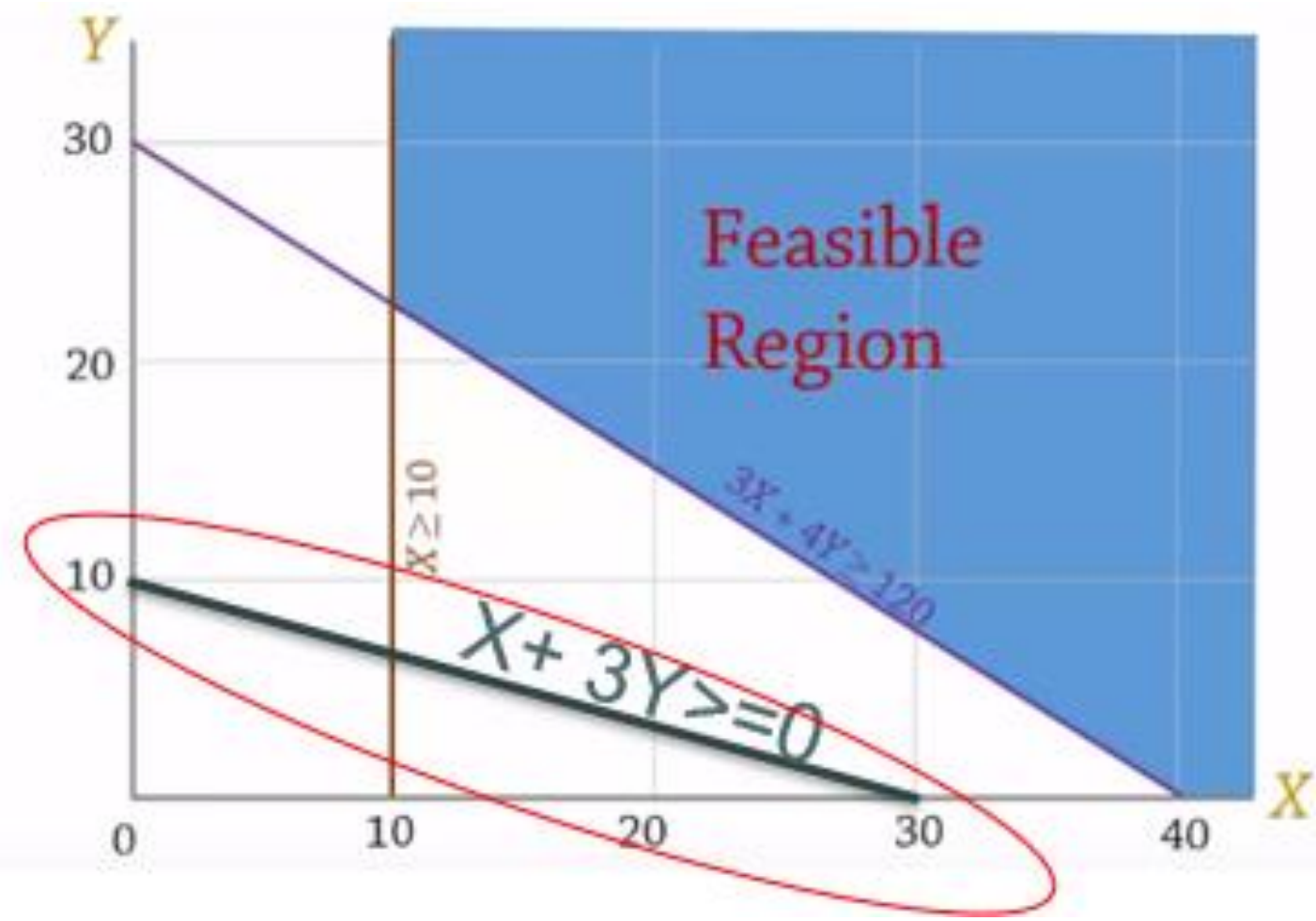
s.t.

$$X + 3Y \geq 30$$

$$3X + 4Y \geq 120$$

$$X \geq 10$$

$$X, Y \geq 0$$



Ex: Prisoner's Dilemma

- Two suspects arrested for a crime
- Prisoners decide whether to confess or not to confess
- If both confess, both sentenced to 3 months of jail
- If both do not confess, then both will be sentenced to 1 month of jail
- If one confesses and the other does not, then the confessor gets freed (0 months of jail) and the non-confessor sentenced to 9 months of jail
- What should each prisoner do? Prepare a payoff matrix

Prisoner's Dilemma

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- What should each prisoner do?

Prisoner 2

Prisoner 1

	Confess	Not Confess
Confess	-3,-3	0,-9
Not Confess	-9,0	-1,-1

Battle of gender interests

- A couple deciding how to spend the evening
- Wife would like to go for a movie, gain for wife is 2 but the gain for husband is 1
- Husband would like to go for a cricket match, then the gain for husband is 2 the gain for wife is 1
- Both however want to spend the time together
- Scope for strategic interaction

Husband

Wife

	Movie	Cricket
Movie	2,1	0,0
Cricket	0,0	1,2

Bidding up to 50

- Two-person game
- Start with a number from 1-4
- You can add 1-4 to your opponent's number and bid that
- The first person to bid 50 (or more) wins
- Game theory tells us that **person 2** always has a winning strategy…. Let us try
  - Bid 5, 10, 15, …, 50

Examples of two-person zero-sum games:

(1) Enterprise bargaining (a union negotiating a new contract with management)
(2) US and Japan participating in war games
(3) Two political parties (Labour and Liberals) competing against each other- one wants secure passage of legislation and the other wants to defeat the proposed legislation.
(4) A retailer tries to improve its market share with a new product and the competitor tries to minimise its losses in the same market.
(5) A company bidding for a large scale contract with the government

Suppose there are two manufacturers of cigarettes: Philip and Baltimore. Their profit (in millions of $) when they advertise and when they do not advertise, are shown in the following table.

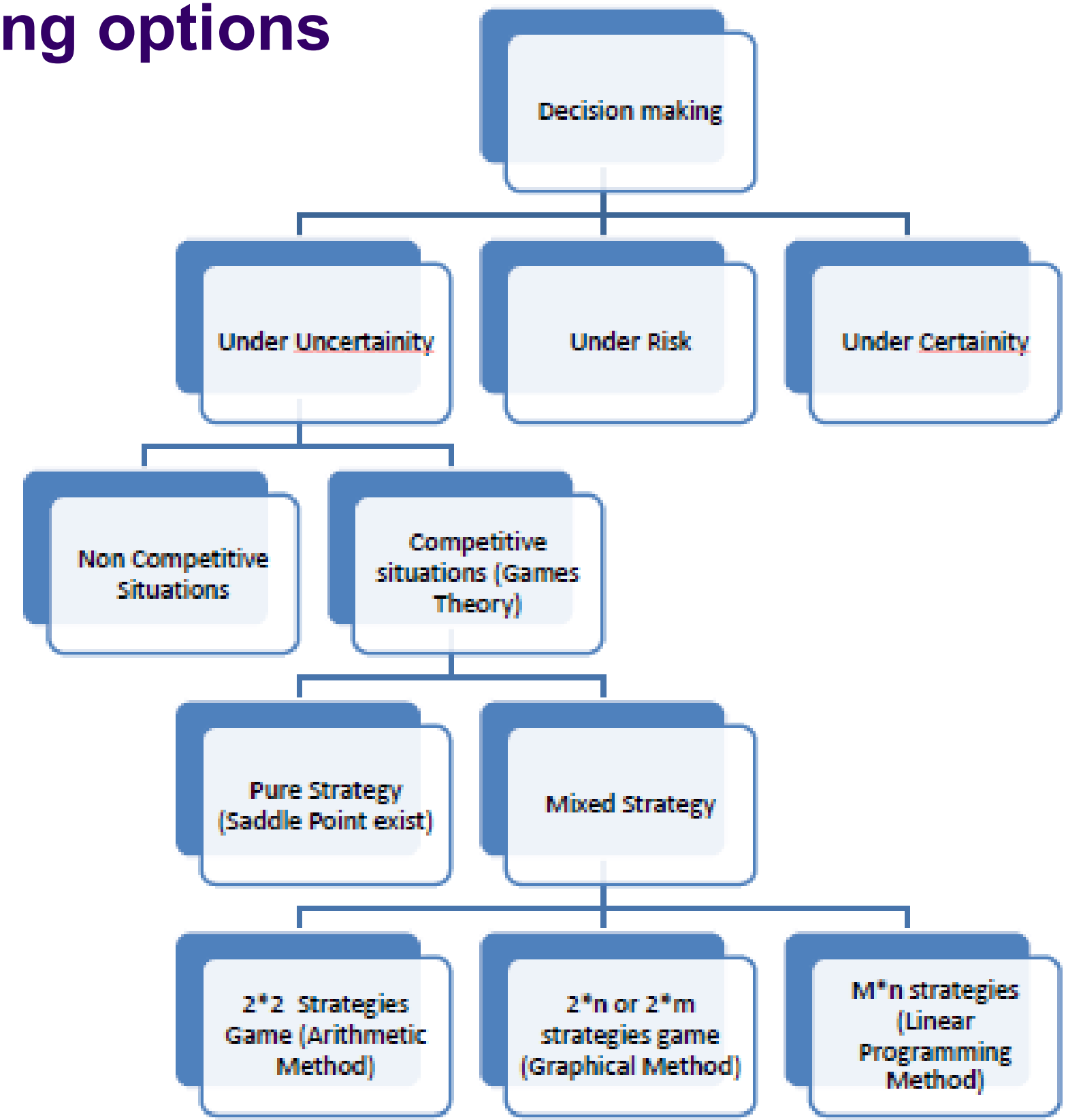
Not a zero sum game

Baltimore's action	Philip's action	
	Advertise	Do not advertise
Advertise	25, 25	60, 20
Do not advertise	20, 60	45, 45

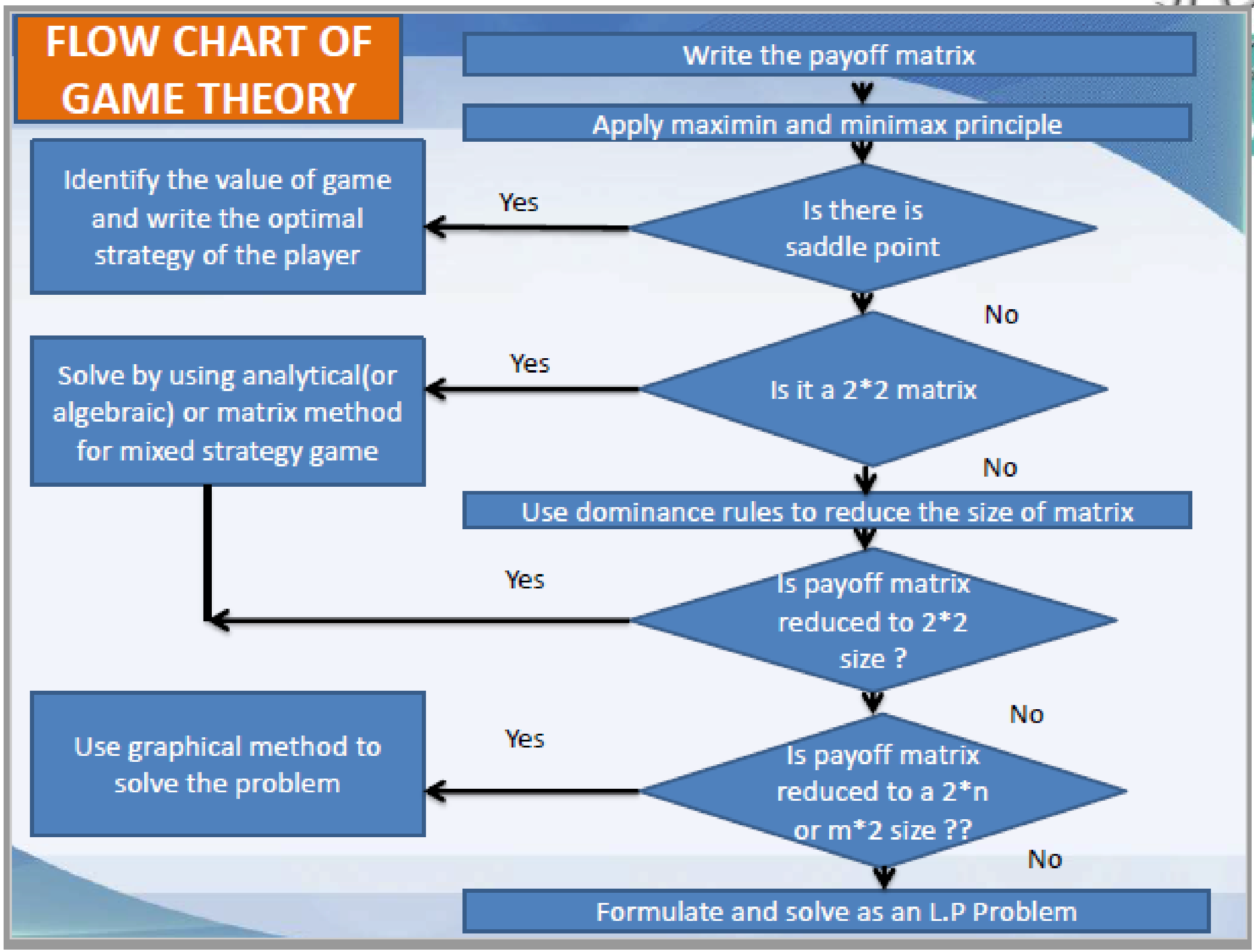
Mixed strategies

- When saddle point do not exists the best strategy is a mix strategy and the value of the game lies between the best values obtained for player and opponent.
- In mixed strategies Value of the game lies between the two values of Maximin and Mini Max
- If the matrix is bigger than 2*2 use the **dominance strategy** to reduce the size of the matrix
- There are 3 ways of finding this best strategy. They are,
- 1) if the payoff matrix is 2*2 then use **oddment or simple probability** theory
- 2) If the payoff matrix is **2*n or m*2 then use the graphical method-**
  Which can be used only if the player has only two available decisions.
- 3) if the payoff matrix is **m*n use x Simplex Method**
  Which can be used independent of the number of decisions available to the player.

Decision making options



FLOW CHART OF GAME THEORY



Homework on Game Theory

Ex 1
Check for dominance and find the Nash equilibrium point/saddle point value in

Exercise 1.1		Opponent	
		B1	B2
PLAYER	A1	7	8
	A2	4	5
	A3	3	1

Exercise 1.3		Opponent			
		B1	B2	B3	B4
PLAYER	A1	6	4	5	7
	A2	2	3	1	0

Exercise 1.2		Opponent		
		B1	B2	B3
PLAYER	A1	3	-1	2
	A2	-4	-1	13
	A3	2	-2	-1

Ex 1
Check for dominance and find the Nash equilibrium point/saddle point value in

Exercise 1.1

		Opponent	
		B1	B2
PLAYER	A1	7	8
	A2	4	5
	A3	3	1

Exercise 1.3

		Opponent			
		B1	B2	B3	B4
PLAYER	A1	6	4	5	7
	A2	2	3	1	0

Exercise 1.2

		Opponent		
		B1	B2	B3
PLAYER	A1	3	-1	2
	A2	-4	-1	13
	A3	2	-2	-1

- Ex 2
- Check for dominance and find the Nash equilibrium point/saddle point value in respect of the following gain matrix.

Ex 2.1		Opponent			
		B1	B2	B3	B4
PLAYER	A1	7	2	8	5
	A2	9	5	3	1
	A3	8	3	6	10
	A4	3	1	4	3

Exercise 2.2		Opponent				
		B1	B2	B3	B4	B5
PLAYER	A1	4	6	5	10	6
	A2	7	8	5	9	10
	A3	8	9	11	10	9
	A4	6	4	10	6	4

Exercise 2.3

		Opponent			
		B1	B2	B3	B4
PLAYER	A1	5	4	3	7
	A2	3	2	1	4
	A3	9	1	2	8
	A4	7	2	10	3

Ex 3:

a)In a game between two players A and B, the player A chooses a number S from the set -1, 0, 1 and his opponent B chooses a number T form the same set. Assume that when one chooses, the chosen number is put back. After the choices have been disclosed B pays A the amount given by S*(T-S)+T*(T+S). If the payment made by B to A is negative, then that implies that A pays that amount to B. This is a zero-sum game. **Construct the pay-off matrix for player A and Player B**

(a) Construct the pay-off matrix for player B

S	T	**S*(T-S)+T(T+S)**
-1	-1	**2**
-1	0	**-1**
-1	1	**-2**
0	-1	**1**
0	0	**0**
0	1	**1**
1	-1	**-2**
1	0	**-1**
1	1	**2**

(a) In a game between two players A and B, the player A chooses a number S from the set -1, 0, 1 and his opponent B chooses a number T form the same set. Assume that when one chooses, the chosen number is put back. After the choices have been disclosed B pays A the amount given by S*(T-S)+T*(T+S). If the payment made by B to A is negative, then that implies that A pays that amount to B. This is a zero-sum game. **Construct the pay-off matrix for player A and Player B**

(b) Construct the pay-off matrix for player B

(a) Pay-off matrix for player A

		Player B's selection		
		-1	**0**	**1**
Player A's selection	**-1**	2	-1	-2
	0	1	0	1
	1	-2	-1	2

S	T	S*(T-S)+T(T+S)
A	B	
-1	-1	**2**
-1	0	**-1**
-1	1	**-2**
0	-1	**1**
0	0	**0**
0	1	**1**
1	-1	**-2**
1	0	**-1**
1	1	**2**

(a) In a game between two players A and B, the player A chooses a number S from the set -1, 0, 1 and his opponent B chooses a number T form the same set. Assume that when one chooses, the chosen number is put back. After the choices have been disclosed B pays A the amount given by S*(T-S)+T*(T+S). If the payment made by B to A is negative, then that implies that A pays that amount to B. This is a zero-sum game. **Construct the pay-off matrix for player A and Player B**

(b) Construct the pay-off matrix for player B

(a) Pay-off matrix for player A

		Player B's selection		
		-1	**0**	**1**
Player A's selection	**-1**	2	-1	-2
	0	1	0	1
	1	-2	-1	2

(b) Pay-off matrix for player B

		Player A's selection		
		-1	**0**	**1**
Player B's selection	**-1**	-2	-1	2
	0	1	0	1
	1	2	-1	-2

S	T	S*(T-S)+T(T+S)
A	B	
-1	-1	2
-1	0	-1
-1	1	-2
0	-1	1
0	0	0
0	1	1
1	-1	-2
1	0	-1
1	1	2

Ex 4:A banker shows you three boxes, with removable bottoms, labelled 1, 2 and 4 and containing each as many dollar notes as the label says. He tells you that he will remove one of the bottoms, unknown to you, and invites you to put into two of the boxes as a dollar notes indicated by their labels. **He also puts the same dollar notes as you did.** You then open these two boxes and get those dollar notes that are still there. This is a zero-sum game.

(a)Construct the pay-off matrix for YOU
(b)Construct the pay-off matrix for banker
(c)Can this game be solved through IESDS solution concept?

- A banker shows you three boxes, with removable bottoms, labelled 1, 2 and 4 and containing each as many dollar notes as the label says. He tells you that he will remove one of the bottoms, unknown to you, and invites you to put into two of the boxes as a dollar notes indicated by their labels. He also puts the same dollar notes as you did. You then open these two boxes and get those dollar notes that are still there. Show that this is a zero-sum game.

(a) Pay-off matrix for player "YOU"

		Banker's selection		
		Remove 1	**Remove 2**	**Remove 4**
Your selection	**Put in 1 and 2**	1	-1	3
	Put in 1 and 4	3	5	-3
	Put in 2 and 4	6	2	-2

- A banker shows you three boxes, with removable bottoms, labelled 1, 2 and 4 and containing each as many dollar notes as the label says. He tells you that he will remove one of the bottoms, unknown to you, and invites you to put into two of the boxes as a dollar notes indicated by their labels. He also puts the same dollar notes as you did. You then open these two boxes and get those dollar notes that are still there. Show that this is a zero-sum game.

(a) Pay-off matrix for player "YOU"

		Banker's selection		
		Remove 1	**Remove 2**	**Remove 4**
Your selection	**Put in 1 and 2**	1	-1	3
	Put in 1 and 4	3	5	-3
	Put in 2 and 4	6	2	-2

(b) Pay-off matrix for player "Banker"

		Your selection		
		Put in 1 and 2	**Put in 1 and 4**	**Put in 2 and 4**
Banker's selection	**Remove 1**	-1	-3	-6
	Remove 2	1	-5	-2
	Remove 4	-3	3	2